IGCSE 0607 Practice Questions — E2.6

Inequalities

Total Questions: 6 — **Total Marks:** 33 Marks

Table of Contents & Quick Navigation

- **E2.6** Inequalities
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

---

1 **(a)** Solve the inequality.

$$4x - 3 \leq 7x + 12$$

..... **[2]**

(b) Solve the inequality.

$$-8 < 2x - 1 \leq 7$$

..... **[2]**

(c) Write down the largest integer and the smallest integer that satisfy the inequality in part (b).

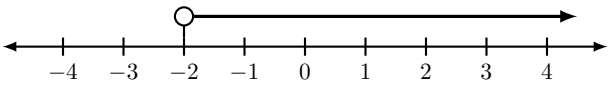
Largest = ... , Smallest = ... **[2]**

2 **(a)** Solve the inequality.

$4x - 3 < 2x + 9$

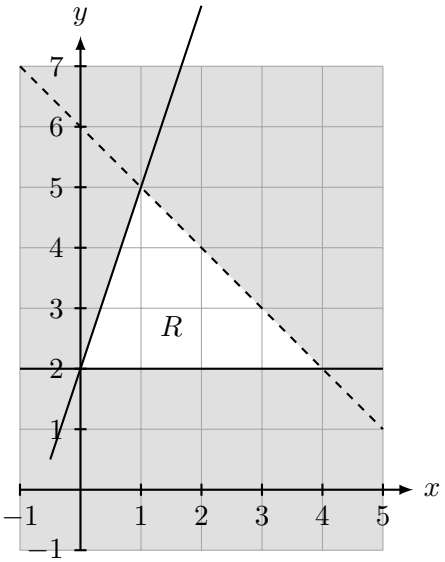
..... **[2]**

(b) Write down the inequality shown on the number line.



..... **[2]**

3 A region R is defined by three inequalities.



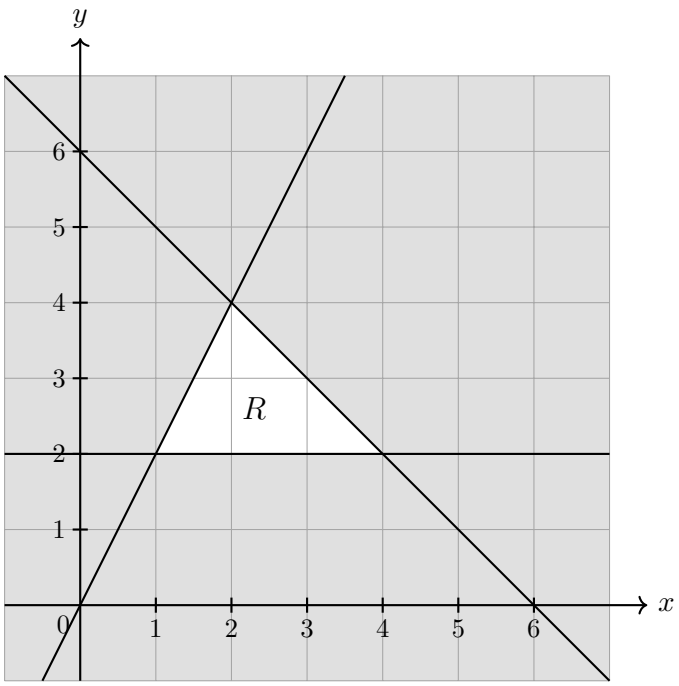
The region R, unshaded, is defined by three inequalities.
Write down these three inequalities.

.....

.....

..... **[6]**

4 The diagram shows a region R unshaded.



Write down the three inequalities that define the region R.

.....

.....

..... **[3]**

5 Tickets for a community show cost \$5 for children and \$10 for adults.
The maximum number of tickets that can be sold is 500.
The organizers must sell at least 150 adult tickets.
The total amount of money raised from ticket sales must be at least \$3000.

Let x be the number of children's tickets sold and y be the number of adult tickets sold.

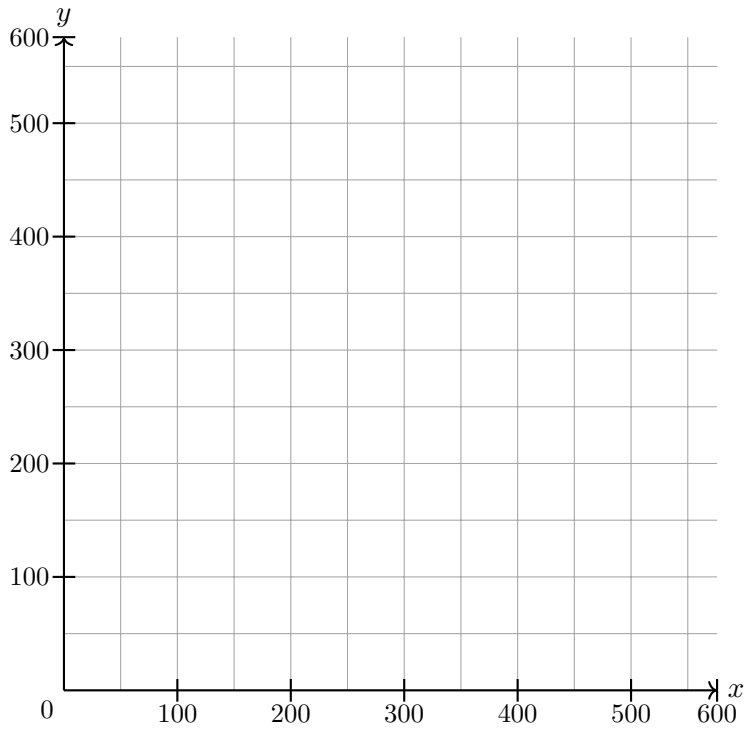
(a) Write down three inequalities in x and y to show this information.

.....

.....

..... **[3]**

(b) On the grid, draw three lines to show these inequalities.
Shade the unwanted regions to leave the region representing the possible number of tickets sold.



[4]

(c) The organizers want to maximize the number of children's tickets sold.
Find this maximum number of children's tickets and the corresponding number of adult tickets.

Children's tickets = ... , Adult tickets = ... **[2]**

6 **(a)** Solve the double inequality.

$$-7 \leq 3x - 1 < 11$$

..... **[3]**

(b) Write down all the integer values of x that satisfy the inequality in part (a).

..... **[2]**